KAKATIYA OF INSTITUTE OF TECHNOLOGY & SCIENCE WARANGAL – 15

DEPARTMENT OF ELECTRONICS & COMMUNICATIONS ENGINEERING

A

Seminar Presentation

On

"Multifunctional Smart Stick for Visually Impaired with Obstacle Detection"

By

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Abstract

Visual impairment is one of the biggest challenges that are faced by millions of people across the world on a day-to-day basis. According to a report by the World Health Organization, a significant number of people are either partially blind or completely blind, depending upon others for safe passage. The project that has been been nominated is Enhanced Smart Blind Stick. The proposed system relies on ultrasonic sensors to identify barriers in front of the person and gives feedback in terms of vibration and voice messages. Moreover, the system is improved with an alert mechanism for water and pits to avoid life threats like slipping over water or stepping on pits. For emergency conditions, GPS and GSM modules are added in the system for sending the location of the person to a contact already saved in the system. This smart stick is very easy to use, cost-effective, portable, and consumes low power. It greatly boosts the safety, confidence, and level of independence that the visually impaired have, and can be further enhanced with technologies such as AI-powered object recognition, and so on.

Introduction

Millions around the globe are struggling with visual disabilities and impaired eyesight. Conventional white canes help to locate only obstacles that directly impede the user, and they do not warn users of hazards like water, holes in the ground, or moving obstacles. Other assistive options fail to sense human or animal activity that is occurring directly in front of the user. This project seeks to develop a multi-functional smart stick using Arduino and other advanced sensor devices that will detect obstacles both static (e.g., walls) as well as dynamic (moving objects like humans and animals) and give immediate notification to the user through vibration and verbal instructions. In addition, it includes a GPS and GSM-based emergency response feature that can alert a specified friend/relative of the user's current location. In conclusion, it is proposed that the Smart Stick will provide enhanced user safety, more intelligent detection, and improved user assistance compared to existing assistive devices.

Challenges

- Detecting obstacles in front of the user.
- Identifying water on the road to avoid slipping.
- Detecting pits and uneven surfaces to prevent falling.
- Detecting moving objects such as humans and animals.
- Avoiding sudden accidents due to unexpected movements.
- Sending live location to a saved contact using GPS and GSM.
- Making the system portable and easy to carry.
- Keeping the system low cost and affordable.

Existing Method

- Traditional white cane can detect only very close obstacles and gives very less reaction time.
- Existing methods cannot detect water, pits, or uneven surfaces, which may cause slipping or falling.
- They cannot detect moving objects such as humans or animals, leading to sudden accidents.
- Most systems do not provide any emergency communication or location sending facility.
- Some available smart devices are costly, complex, and not user-friendly.
- Overall, existing methods do not provide complete safety, reliability, and independence to visually impaired people.

Proposed Method & Justification

➢**Proposed Method:**

•Use **Arduino** as controller.

•Use **Ultrasonic sensor** to detect obstacles.

•Use **water sensor** to detect water and pits.

•Use **GPS & GSM** to send location in emergency.

•Use **buzzer/vibration motor & voice module** for alerts.

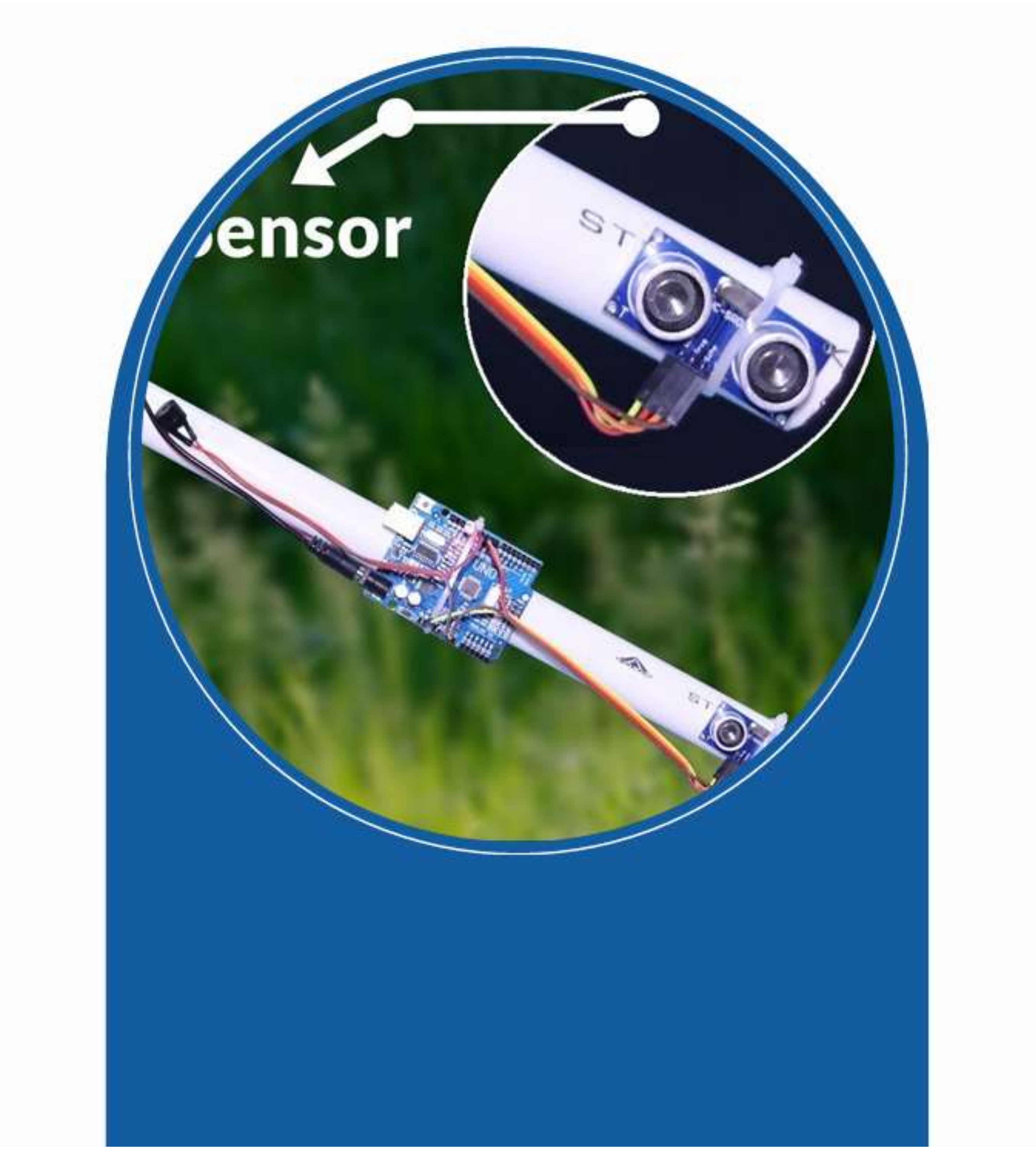
➢**Justification:**

•Low cost and easy to implement.

•Works in real-time.

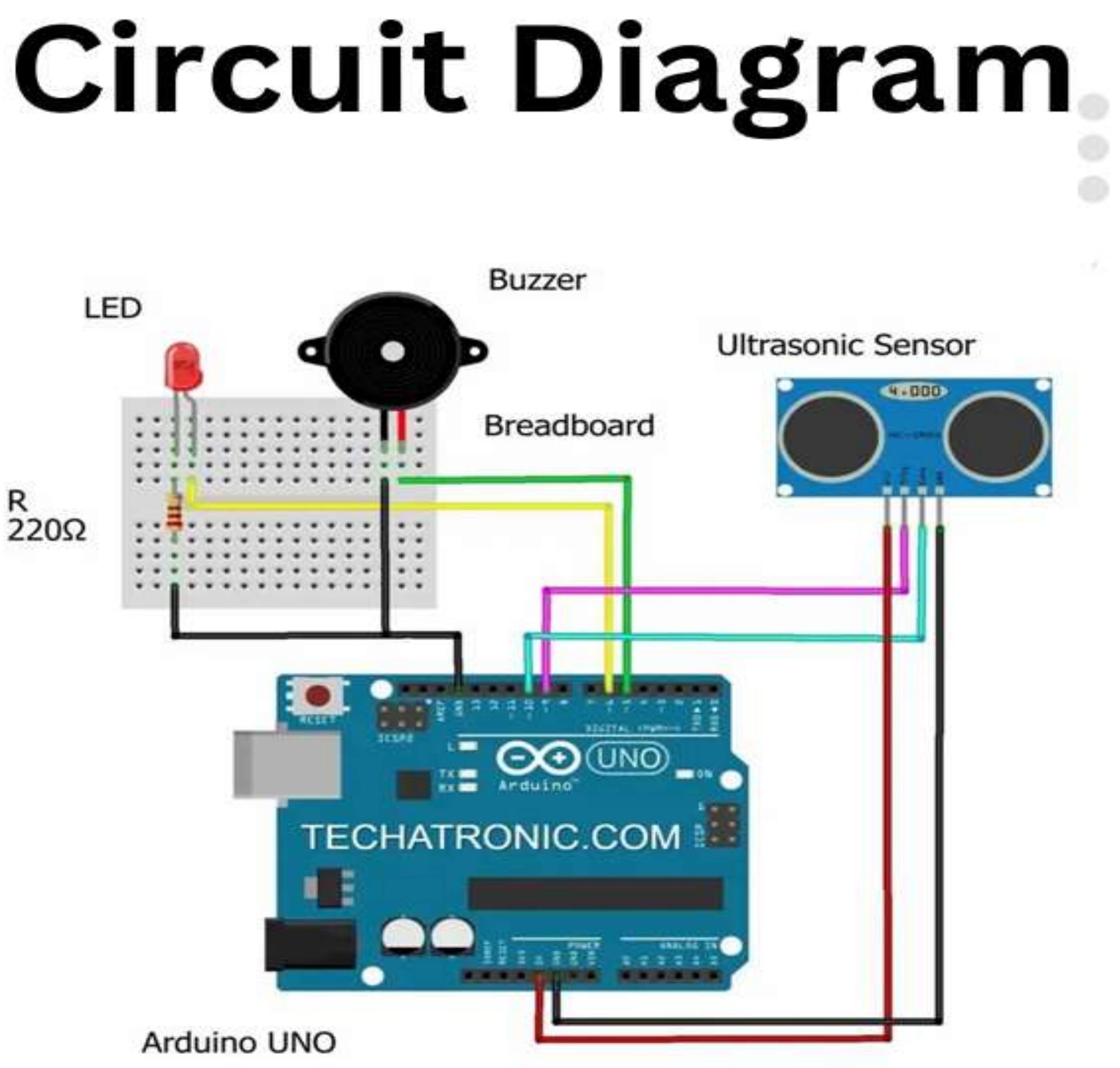
•Portable and low power consumption.

•Highly useful for blind people.



Circuit Diagram Overview

The circuit uses Arduino UNO as the main controller to manage all components. An ultrasonic sensor is used to detect obstacles in front of the user. When an obstacle is detected, Arduino processes the signal and activates the buzzer and LED to give an alert. This simple circuit forms the basic obstacle detection unit of the smart stick system.



Advantages

- ➢Easy to use.
- ➢Low cost.
- ➢Portable.
- ➢Low power consumption.
- ➢Real-time obstacle detection.
- ➢Emergency support using GPS & GSM.
- ➢Increases safety and confidence of blind people.

Limitations

- ➢Cannot detect very small or transparent objects.
- ➢GPS may not work properly inside buildings.
- ➢Needs battery charging.
- ➢Performance may vary in extreme weather conditions.

Applications

- For visually impaired people to walk safely and independently.
- On roads and footpaths to avoid obstacles, water, and pits.
- In crowded places to detect moving people and animals.
- In public places like railway stations, bus stands, and markets.
- In hospitals, colleges, schools, and offices.
- During night time or low visibility conditions.
- For emergency situations to send location to family members.
- In unknown or new environments for safe navigation.

Components

➢**Software:**
•Arduino IDE
•Serial Monitor
•Python (optional)
➢**Hardware:**
•Arduino UNO / Nano
•Ultrasonic Sensor (HC-SR04)
•GPS Module (NEO-6M)
•GSM Module (SIM800L / SIM900)
•Buzzer / Vibration Motor
•Battery Pack
•Jumper Wires
•Stick Body

Conclusion

The smart stick with multiple capabilities for those who cannot see will serve as a very effective solution to enhance their safety and independence. In addition to the basic functions of a traditional smart stick, this professional version will also include obstacle detection, detection of water, detection of holes in the ground, and detection of moving objects (e.g. people or animals), along with providing alerts via vibration and voice commands. Adding GPS/GSM technology will help to make the smart stick more reliable, as it allows users of the smart stick to send their exact location to someone when they need help. Ultimately, this project will help individuals who are visually impaired to feel more comfortable and safe in their day-to-day lives.

References

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THANK YOU